

SHOPPING AND BILLING SYSTEM

Python Programming Project Statement

Project Type	First-Year Python Project
Programming Language	Python
Application Type	Menu-Driven Shopping and Billing Simulation
Development Platform	Google Colab / Python IDE

Project Summary

The Shopping and Billing System is a beginner-friendly Python application developed to simulate the billing counter of a small shop. The program allows users to view a product list, add products with quantities to a cart, view the cart and generate a final bill through a simple menu-driven interface.

The bill applies a quantity-based discount and 5% GST automatically. The project focuses on applying fundamental Python programming concepts to a real-life problem. It does not connect to a real payment system, database or inventory; it is intended only for academic learning and demonstration.

Submitted as a Python Programming Project

1. Introduction

Billing software is a common example of a computer-based information system. Shops use it to identify products, calculate the cost of each item, apply offers and taxes, and print a correct bill. This project creates a simplified version of such a system using Python.

The program is designed around a menu. The user selects an operation, provides the required information, and receives an appropriate result. After completing an operation, the menu appears again until the user chooses to exit.

2. Problem Statement

Design and implement a Python program that can display a list of products, let a customer add products to a cart, and generate a bill. The system should calculate the line total of each item, apply a discount based on the purchase amount, add GST, and display the final amount payable.

3. Objectives

- To develop a simple real-life application using Python.
- To understand how dictionaries can store structured product information.
- To use lists to hold the items in a shopping cart.
- To use functions to divide the program into manageable modules.
- To apply conditional statements for decision making, such as discount slabs.
- To use loops for repeated menu operations.
- To perform arithmetic operations for totals, discount and tax.

4. Scope of the Project

The current project is limited to a fixed product catalogue, a single cart and in-memory data storage. It is suitable for academic demonstration and learning. It can later be extended with file handling, databases, customer details, stock management and a graphical user interface.

5. System Features and Working

5.1 Display Products

The program lists all ten products with their product number, name and price in rupees.

5.2 Add Product to Cart

The user chooses a product number and enters a quantity. The program checks that the product exists, calculates the line total (price multiplied by quantity) and adds the item to the cart. An unknown product number gives an error message.

5.3 View Cart

The program shows each item in the cart with its price, quantity and total, followed by the subtotal. If the cart has no items, it reports that the cart is empty.

5.4 Generate Bill

The program prints every item and calculates the subtotal. A discount of 5% is given for a subtotal of Rs. 500 or more, and 10% for Rs. 1000 or more. GST of 5% is then added on the discounted amount, and the final amount is displayed. A bill cannot be generated for an empty cart.

5.5 Exit

The program ends with a thank-you message.

6. Program Flow

Step	Operation
1	Start the program
2	Display the main menu
3	Accept the user's choice
4	Perform the selected operation
5	Validate the product number or cart where required
6	Display the result
7	Return to the main menu
8	Exit when the user selects Exit

Main Menu

```
=====
      SHOPPING & BILLING
=====
1. Display Products
2. Add Product to Cart
3. View Cart
4. Generate Bill
5. Exit
=====
```

7. Python Concepts Used

Functions

Separate functions are used for displaying products, adding to the cart, viewing the cart and generating the bill. This makes the program easier to read and maintain.

Dictionary and List

A dictionary named 'products' stores product numbers as keys, with the name and price as values. A list named 'cart' stores the items the customer has chosen.

Conditional Statements

if, elif and else statements are used to select menu options, validate the product number and choose the discount slab.

Loops

A while loop keeps the menu running until the user chooses the Exit option. A for loop is used to display the products and the cart items.

Input and Arithmetic Operations

The input() function accepts information from the user. Arithmetic operators are used to calculate line totals, the discount, GST and the final amount.

8. Sample Test Cases

Test	Input	Expected Result
Add product	Rice (1), quantity 5	Line total Rs. 300
Add product	Milk (4), quantity 2	Line total Rs. 60
View cart	Option 3	Subtotal Rs. 360
Bill, no discount	Option 4	Discount 0, GST 18.0, Final 378.0
Bill, 5% discount	Shampoo x5 (Rs. 600)	Final Amount 598.5
Bill, 10% discount	Shampoo x10 (Rs. 1200)	Final Amount 1134.0
Invalid product	Product number 99	Invalid product number.
Empty cart bill	Option 4 with empty cart	Cart is empty. Add products first.

9. Advantages

- Simple and beginner-friendly interface.
- Uses important Python programming concepts.
- Easy to modify and expand.
- Calculates discount and GST automatically, avoiding manual errors.
- Provides clear messages for common mistakes.

10. Limitations

- Data is stored only while the program is running.
- The product list is fixed inside the program.
- Non-numeric input for product number or quantity is not handled.
- Items cannot be removed from the cart.
- The application cannot process real payments.
- The project is intended for educational use only.

11. Future Enhancements

- Add try/except handling and quantity validation.
- Add options to remove items and clear the cart.
- Save bills permanently in a file or database.
- Add customer details, bill number and date on the bill.
- Add stock management for products.
- Add payment options.
- Develop a graphical user interface using Tkinter.

12. Learning Outcomes

This project provides practical experience in converting a real-life requirement into a structured Python program. It improves understanding of functions, dictionaries, lists, loops, conditional statements, input handling and basic problem solving.

13. Conclusion

The Shopping and Billing System successfully demonstrates how fundamental Python programming concepts can be combined to create a useful menu-driven application. The system performs the core steps of a billing counter: showing products, adding items to a cart, viewing the cart and generating a bill with discount and GST.

Although the project is a simplified simulation and does not provide the validation, persistence or payment features of a real shop system, it provides a strong foundation for future development.

Project Status: Completed – Educational Python Simulation

End of Project Statement